S. Meng (2023) "Log Calabi–Yau Structure of Projective Threefolds Admitting Polarized Endomorphisms," International Mathematics Research Notices, Vol. 2023, No. 24, pp. 21272–21289
Advance Access Publication December 8, 2022
https://doi.org/10.1093/imrn/rnac308

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Log Calabi–Yau Structure of Projective Threefolds Admitting Polarized Endomorphisms

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Dedicated to Professor De-Qi Zhang on his sixtieth birthday

Let X be a normal projective variety admitting a polarized endomorphism f, that is, $f^*H \sim qH$ for some ample divisor H and integer $q > 1$. It was conjectured by Broustet and Gongyo that X is of Calabi–Yau type, that is, (X, Δ) is lc for some effective $\mathbb{Q}$-divisor such that $K_X + \Delta \sim_{\mathbb{Q}} 0$. In this paper, we establish a general guideline based on the equivariant minimal model program and the canonical bundle formula. In this way, we prove the conjecture when X is a smooth projective threefold.

1 Introduction

We work over an algebraically closed field of characteristic 0. Let $f : X \to X$ be a surjective endomorphism of a normal projective variety X. In the curve case, it is well known by the Hurwitz formula that X is either a rational curve or an elliptic curve when f is non-isomorphic. This is equivalent to saying that the anticanonical divisor $-K_X$ is effective. In higher-dimensional case, to easily eliminate the distraction of automorphism, one focuses on the *polarized* endomorphism f, that is, $f^*H \sim qH$ for some ample divisor H and integer $q > 1$. Then, by making use of the ramification divisor

Received March 24, 2022; Revised October 9, 2022; Accepted October 18, 2022
Communicated by Prof. Laura DeMarco

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formula, Zhang and the author showed that $-K_X$ is effective when X is $\mathbb{Q}$-Gorenstein. However, the effectivity of the anticanonical divisor, though important, can say very few on the detailed characterization of higher-dimensional varieties.

A delicate operation is running the f-equivariant (after iteration) minimal model program. The smooth surface case is settled by Nakayama and the general higher-dimensional situation is settled by Zhang and the author (cf. [3, 13–15, 17]). In this way, Broustet and Gongyo [2] proposed the following conjecture and proved the surface case. Recall that a normal projective variety X is of *Calabi–Yau type* if (X, Δ) is an lc pair for some effective Weil $\mathbb{Q}$-divisor Δ such that $K_X + \Delta \sim_{\mathbb{Q}} 0$. By the abundance (cf. [7, Theorem 1.2]), the latter condition is equivalent to $K_X + \Delta \equiv 0$. We also call the pair (X, Δ) *log Calabi–Yau*.

Conjecture 1.1. Let X be a normal projective variety admitting a polarized endomorphism. Then X is of Calabi–Yau type.

In higher-dimensional cases, Conjecture 1.1 has been recently verified for rationally connected smooth projective varieties by Yoshikawa; see [20] or Theorem 2.8. The main purpose of this paper is to give a partial guideline on Conjecture 1.1 and to provide a full solution for the case of smooth projective threefolds. The following is our main result.

Theorem 1.2. Let X be a smooth projective threefold admitting a polarized endomorphism f. Then X is of Calabi–Yau type, that is, (X, Δ) is an lc pair for some effective $\mathbb{Q}$-divisor Δ with $K_X + \Delta \sim_{\mathbb{Q}} 0$.

We briefly explain the strategy and difficulty in our proof. We first run the f-equivariant minimal model program that ends up with a Q-abelian variety Y, that is, a quasi-étale quotient of an abelian variety. Since the smooth rationally connected case has been verified by Yoshikawa, we may assume $\dim(Y) > 0$. Then we observe the fibration $\pi : X \to Y$ and its f-periodic general fibre. By applying the canonical bundle formula and ramification divisor formula, the natural idea is to reduce the problem to the following Conjecture 1.3 proposed by Yoshinori Gongyo.

Conjecture 1.3 (Gongyo). Let $f : X \to X$ be a q-polarized endomorphism of a smooth projective variety X. Then $(X, \frac{R_f}{q-1})$ is an lc pair after iteration.

However, for the surface case, we cannot fully prove Conjecture 1.3 for $\mathbb{P}^2$, which is the only left case. So the main difficulty of proving Theorem 1.2 remains in the case

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when $\pi : X \to Y$ is a $\mathbb{P}^2$-bundle over an elliptic curve Y. By applying the Iitaka fibration of the anticanonical divisor, we only need to focus on the case when $-K_X$ is big; see Section 5. This new condition allows us to reduce the problem to the only very concrete case when $X \cong \mathbb{P}_Y(\mathcal{F}_2 \oplus \mathcal{L})$ where $\mathcal{F}_2$ is the unique indecomposable rank 2 vector bundle with nontrivial global sections and $\mathcal{L}$ is a line bundle of negative degree.

2 Preliminaries

We use the following notation throughout this paper.

Notation 2.1. Let X be a projective variety.

- The symbols $\sim$ (resp. $\sim_{\mathbb{Q}}$, $\equiv$) denote the *linear equivalence* (resp. $\mathbb{Q}$-*linear equivalence*, *numerical equivalence*) on $\mathbb{Q}$- (or $\mathbb{R}$-) Cartier divisors. We also use $\equiv$ to denote the *numerical equivalence* of 1-cycles on X.
- Denote by $\mathrm{NS}(X) = \mathrm{Pic}(X)/\mathrm{Pic}^0(X)$ the *Néron–Severi group* of X. Let $\mathrm{N}^1(X) := \mathrm{NS}(X) \otimes_{\mathbb{Z}} \mathbb{R}$ the space of $\mathbb{R}$-Cartier divisors modulo numerical equivalence and $\rho(X) := \dim_{\mathbb{R}} \mathrm{N}^1(X)$ the *Picard number* of X. Let $\mathrm{N}_1(X)$ be the dual space of $\mathrm{N}^1(X)$ consisting of 1-cycles. Denote by $\mathrm{Nef}(X)$ the cone of *nef divisors* in $\mathrm{N}^1(X)$ and $\overline{\mathrm{NE}}(X)$ the dual cone consisting of *pseudo-effective 1-cycles* in $\mathrm{N}_1(X)$.
- Let $f : X \to X$ be a surjective endomorphism. A subset $Y \subseteq X$ is f^{-1}-*invariant* (resp. f^{-1}-*periodic*) if $f^{-1}(Y) = Y$ (resp. $f^{-s}(Y) = Y$ for some $s > 0$).
- A surjective endomorphism $f : X \to X$ is *q-polarized* if $f^*H \sim qH$ for some ample Cartier divisor H and integer $q > 1$; see [13, Propositions 1.1] for the equivalent definitions.
- A smooth projective variety X is *rationally connected* if any two general points of X can be connected by a chain of rational curves.
- A normal projective variety X is of *Fano type*, if there is an effective Weil $\mathbb{Q}$-divisor Δ on X such that the pair (X, Δ) has at worst klt singularities and $-(K_X + \Delta)$ is ample and $\mathbb{Q}$-Cartier. If $\Delta = 0$, we say that X is a *(klt) Fano variety*.
- Let Y be a projective variety and $\mathcal{E}$ a vector bundle of rank n. Denote by $\pi : \mathbb{P}_Y(\mathcal{E}) \to Y$ the projective bundle of hyperplanes in $\mathcal{E}$ (not lines in $\mathcal{E}$), so that $\pi_* \mathcal{O}_{\mathbb{P}_Y(\mathcal{E})}(1) = \mathcal{E}$.

The following two lemmas are well known and useful.

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Lemma 2.2. Let $\pi : X \to Y$ be a birational morphism of two normal projective varieties. Then Y is of Calabi–Yau type if X is of Calabi–Yau type.

Proof. Suppose the pair (X, Δ_X) is log Calabi–Yau. Let $\Delta_Y := \pi_* \Delta_X$. Then $K_Y + \Delta_Y = \pi_*(K_X + \Delta_X) \sim_{\mathbb{Q}} 0$. Note that $K_X + \Delta_X = \pi^*(K_Y + \Delta_Y)$. By [11, Lemma 3.38], (Y, Δ_Y) has singularities not worse than (X, Δ_X). So (Y, Δ_Y) is lc. ■

Lemma 2.3. Let $\pi : X \to Y$ be a quasi-étale finite surjective morphism of normal projective varieties. Then X is of Calabi–Yau type if and only if so is Y.

Proof. Suppose (Y, Δ_Y) is log Calabi–Yau. Let $\Delta_X = \pi^* \Delta_Y$. Since π is quasi-étale, $K_X = \pi^* K_Y$ and hence (X, Δ_X) is log Calabi–Yau by [11, Proposition 5.20].

Conversely, assume that (X, Δ_X) is log Calabi–Yau. Note that the Galois closure of π is still quasi-étale by [8, Theorem 3.7]. So we may assume that π is the quotient map of X by a finite group G. Let $\Delta = \frac{1}{|G|} \sum_{g \in G} g^* \Delta_X$ and $\Delta_Y = \frac{1}{\deg \pi} \pi_* \Delta$. Then $\Delta = \pi^* \Delta_Y$. Let $\phi : W \to X$ be a log resolution of (X, Δ) and E a ϕ-exceptional prime divisor. Then we have the discrepancy

$$\begin{aligned} a(E, X, \Delta) &= mult_E(K_W - (\phi^* K_X + \frac{1}{|G|} \sum_{g \in G} g^* \Delta_X)) \\ &= \frac{1}{|G|} \sum_{g \in G} mult_E(K_W - (\phi^* K_X + g^* \Delta_X)) \\ &= \frac{1}{|G|} \sum_{g \in G} a(E, X, g^* \Delta_X) \geq -1, \end{aligned}$$

where $mult_E$ means the coefficient of the divisor on E and the last inequality holds by noting that each pair $(X, g^* \Delta_X)$ is log Calabi–Yau for any $g \in G$. Furthermore, we have

$$K_X + \Delta = \frac{1}{|G|} \sum_{g \in G} (K_X + g^* \Delta_X) = \frac{1}{|G|} \sum_{g \in G} g^* (K_X + \Delta_X) \sim_{\mathbb{Q}} 0.$$

So (X, Δ) is log Calabi–Yau. Since π is quasi-étale by the assumption, we have $K_X = \pi^* K_Y$, and hence

$$K_X + \Delta = \pi^* (K_Y + \Delta_Y).$$

Therefore, (Y, Δ_Y) is log Calabi–Yau by [11, Proposition 5.20]. ■

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Proposition 2.4. Let $f : X \to X$ be a polarized endomorphism of a projective variety. Let D be an effective $\mathbb{Q}$-Cartier divisor on X with $\kappa(X, D) = 0$. Then $\operatorname{Supp} D$ is f^{-1}-periodic.

Proof. By [14, Proposition 3.7], we have

$$f^* f_* D \sim_{\mathbb{Q}} f_* f^* D = (\deg f) D.$$

Since $\kappa(X, D) = 0$, we have $f^* f_* D = (\deg f) D$. In particular, $f^{-i} f^i(\operatorname{Supp} D) = \operatorname{Supp} D$ for all $i \geq 0$. By [12, Lemma 8.1], $\operatorname{Supp} D$ is f^{-1}-periodic. ■

A $\mathbb{Q}$-divisor D is said to be $\mathbb{Q}$-*movable* if for any prime divisor Γ, one has $\Gamma \not\subseteq \operatorname{Supp} D'$ for some effective $\mathbb{Q}$-divisor $D' \sim_{\mathbb{Q}} D$.

Let $f : X \to X$ be a polarized endomorphism of a normal projective variety. Denote by T_f the finite union of f^{-1}-periodic prime divisors (cf. [15, Corollary 3.8]). Denote by $P_f := -(K_X + T_f)$.

Proposition 2.5. Let X be a $\mathbb{Q}$-Gorenstein normal projective variety admitting a polarized endomorphism f. Assume further $\kappa(X, -K_X) = 0$. Then $\operatorname{Supp} R_f = T_f$, $-K_X \sim_{\mathbb{Q}} T_f$, and (X, T_f) is lc.

Proof. This follows from [14, Theorem 6.2] and [21, Theorem 1.3]. ■

We generalize [14, Theorem 1.5].

Theorem 2.6. Let $f : X \to X$ be a polarized endomorphism of a $\mathbb{Q}$-factorial normal projective variety. Then $-(K_X + T_f)$ is $\mathbb{Q}$-movable.

Proof. After iteration, we may assume each component of T_f is f^{-1}-invariant. Consider the log ramification divisor formula

$$f^*(-(K_X + T_f)) = -(K_X + T_f) + \Delta_f,$$

where $\Delta_f = R_f - (q-1)T_f$ is effective and contains no common component of T_f. Replacing $-K_X$ by $-(K_X + T_f)$ in the proof of [14, Theorem 6.2], we have that $-(K_X + T_f)$ is effective. Consider the σ-decomposition

$$-(K_X + T_f) \sim_{\mathbb{Q}} P + N,$$

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where $\kappa(X,N)=0$ and P is $\mathbb{Q}$-movable; see [18, Chapter III]. By Proposition 2.4, we may assume each component of N is f^{-1}-invariant after iteration. So $\operatorname{Supp} N \subseteq T_f$. Let Γ be a prime divisor contained in $\operatorname{Supp} N$. By the triangle inequality (cf. [18, Chapter III, Definition 1.1]),

$$\sigma_\Gamma(f^*(-(K_X+T_f))) \le \sigma_\Gamma(-(K_X+T_f)) + \sigma_\Gamma(\Delta_f) = \sigma_\Gamma(-(K_X+T_f)).$$

On the other hand,

$$\sigma_\Gamma(f^*(-(K_X+T_f))) = q\sigma_\Gamma(-(K_X+T_f)).$$

So $\sigma_\Gamma(-(K_X+T_f))=0$ and hence $N=0$. Since P is $\mathbb{Q}$-movable, we may further assume that P has coefficients ≤ 1. ■

We recall the following result by Yoshikawa [20, Proposition 6.2] that Fano type can be preserved by (polarized) equivariant birational map.

Proposition 2.7. Let $f : X \to X$ be a polarized endomorphism of a normal projective variety X. Let $\pi : X \dashrightarrow Y$ be an f-equivariant birational map. Then X is of Fano type if and only if so is Y.

Proof. We may simply take $\Delta=\Gamma=0$ in [20, Proposition 6.2]. ■

We recall the nice result of Yoshikawa [20, Corollary 1.4].

Theorem 2.8. Let X be a rationally connected smooth projective variety admitting a polarized endomorphism. Then X is of Fano type.

3 Singularities of Ramification Divisor

In this section, we give some general results concerning Conjecture 1.3. First, we consider the coefficients of the ramification divisor.

Theorem 3.1. Let $f : X \to X$ be a q-polarized endomorphism of a projective variety. Then, after iteration, the coefficient r_P of R_f on each prime divisor P has $r_P \le q-1$ and the equality holds if and only if P is f^{-1}-periodic.

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Proof. Let b be the number of prime divisors contained in the branch locus of f. Let P be a prime divisor, and denote by $m_P := mult_P$ the coefficient of a divisor on P. Note that f^*P is a reduced divisor when P is not contained in the branch locus of f (in particular, $m_P f^*P = 1$ in this case). So we may set

$$c := \max\{m_P f^*P \mid P \text{ a prime divisor}\},$$

which is a finite number. Choose some $a > 0$ such that $c^b < q^{a/2}$. Choose some $s > 0$ such that $(\frac{q^t}{q^t-1})^{s/t} > c^b$ for any $1 \le t < a$. Let $N = \max\{a, s\}$, and take $n > N$. Let P be a prime divisor that is not f^{-1}-periodic. Let Q be a prime divisor in $f^{-n}(P)$. Let $r = m_P(f^n)^*P$ be the coefficient of $(f^n)^*P$ on Q. We shall show that $r < q^n$.

If P is not f-periodic, then $f^{-i}(P)$ and $f^{-j}(P)$ have no common irreducible component for any $i, j > 0$ with $i \ne j$. Then,

$$r = m_Q(f^n)^*P = \prod_{i=0}^{n-1} m_{f^i(Q)} f^*(f^{i+1}(Q)) \le c^b < q^{a/2} < q^n,$$

where the 1st inequality is because there are at most b terms of $f^i(Q)$ contributing $m_{f^i(Q)} f^*(f^{i+1}(Q)) > 1$. In the following, we consider the case when P is f-periodic with period $t \ge 1$.

Let r_1 be the coefficient of $(f^t)^*P$ on P. Since all irreducible components of $\{f^{-i}(P)\}_{0<i\le t}$ are different with each other, we have $r_1 < c^b < q^{a/2}$. Moreover, we claim that $r_1 < q^t$. Suppose the contrary that we can write $(f^t)^*P = q^tP + E$ for some effective divisor E. By the projection formula,

$$(\deg f^t)P = (f^t)_*(f^t)^*P = (f^t)_*(q^tP + E) = q^t(\deg f^t|_P)P + (f^t)_*E.$$

Note that $\deg f^t = q^{t\cdot\dim(X)}$ and $\deg f^t|_P = q^{t\cdot(\dim(X)-1)}$. Therefore, we have $E = 0$, and hence P is f^{-1}-periodic, a contradiction. So the claim is proved.

Let e be the minimal nonnegative integer such that $f^e(Q) = P$. Let r_2 be the coefficient of $(f^e)^*P$ on Q. Then $r = r_2 \cdot r_1^{(n-e)/t}$. If $t \ge a$, then $r < c^b \cdot (q^{a/2})^{n/a} < q^n$. If $t < a$, then $r < c^b \cdot (q^t - 1)^{n/t} < q^n$. ∎

Remark 3.2. The proof of Theorem 3.1 can be easily applied to consider general surjective endomorphisms, though the statement could be a bit wordy. We leave the readers the pleasure of the coefficient estimate.

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We observe the behaviour of ramification divisors between equivariant birational morphisms.

Proposition 3.3. Let $\pi : X \to Y$ be a birational morphism of $\mathbb{Q}$-factorial normal projective varieties. Let $f : X \to X$ and $g : Y \to Y$ be two q-polarized endomorphisms such that $g \circ \pi = \pi \circ f$. Then,

$$K_X + \frac{R_f}{q-1} = \pi^*(K_Y + \frac{R_g}{q-1}).$$

In particular, $(X, \frac{R_f}{q-1})$ is lc if $(Y, \frac{R_g}{q-1})$ is lc.

Proof. Since K_X and K_Y are $\mathbb{Q}$-Cartier, we may write

$$K_X = \pi^* K_Y + E,$$

where E has supports contained in the exceptional divisor of π. By the ramification divisor formula, we have

$$K_X = f^* K_X + R_f \text{and } K_Y = g^* K_Y + R_g.$$

By the above three equations, we have

$$\pi^* K_Y + E = f^*(\pi^* K_Y + E) + R_f = \pi^* g^* K_Y + f^* E + R_f = \pi^*(K_Y - R_g) + f^* E + R_f.$$

Note that $f^* E = qE$. So we have

$$\pi^* \frac{R_g}{q-1} - \frac{R_f}{q-1} = E.$$

Then we have

$$K_X + \frac{R_f}{q-1} = \pi^*(K_Y + \frac{R_g}{q-1})$$

as desired. ■

The canonical bundle formula plays a key role in our reduction.

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Proposition 3.4. Let $\pi : X \to Y$ be an algebraic fibration where X is an n-dimensional smooth projective variety and Y is normal projective with $K_Y \equiv 0$. Let $f : X \to X$ and $g : Y \to Y$ be q-polarized endomorphisms such that $g \circ \pi = \pi \circ f$ and $f^*K_X \equiv qK_X$. Then $(X, \frac{R_f}{q-1})$ is log Calabi–Yau after iteration if Conjecture 1.3 holds true for $f^s|_F : F \to F$ where F is a f-periodic (of period s) general fibre of π.

Proof. By the ramification divisor formula, we have

$$K_X + \frac{R_f}{q-1} \equiv 0,$$

which holds true after arbitrary iteration.

Let F be a general f-invariant smooth fibre of π after iteration (cf. [5, Theorem 5.1]). Note that $f|_F$ is q-polarized and $R_{f|_F} = R_f|_F$. By the assumption, $(F, \frac{R_{f|_F}}{q-1})$ is lc after iteration. So $(X, \frac{R_f}{q-1})$ is lc over the generic point of π. By the lc canonical bundle formula [6, Theorem 4.1.1], we have

$$0 \equiv K_X + \frac{R_f}{q-1} = \pi^*(K_Y + B + M),$$

where B is effective and M is pseudo-effective. Note that $K_Y \equiv 0$. So $B = 0$, and hence $(X, \frac{R_f}{q-1})$ is lc. ∎

4 Surface Case

We first prove Conjecture 1.3 for surfaces except $\mathbb{P}^2$.

Theorem 4.1. Let $f : X \to X$ be a q-polarized endomorphism of a smooth projective surface with $\rho(X) > 1$. Then $(X, \frac{R_f}{q-1})$ is lc after iteration.

Proof. We apply [13, Theorem 1.8]. If K_X is pseudo-effective, then $R_f = 0$ and we are done. In the following, we may assume K_X is not pseudo-effective. After iteration, we may run an f-equivariant minimal model program of X that ends up with a Fano contraction $\pi : X' \to Y$ with Y being a curve of genus ≤ 1. Then $f^*|_{\mathrm{N}^1(X)} = q\,\mathrm{id}$ and hence $K_X + \frac{R_f}{q-1} \equiv 0$ by the ramification divisor formula. By Proposition 3.3, we may assume $X = X'$. Note that X is a ruled surface over Y. We finish the proof with the following two cases.

Case 1. Suppose Y is elliptic. Note that Conjecture 1.3 holds for curves and $f^*|_{\mathrm{N}^1(X)} = q\,\mathrm{id}$. So we are done by Proposition 3.4.

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Case 2. Suppose $Y = \mathbb{P}^1$. Then $X \cong F_d$ with $d \geq 0$. If $d = 0$, then R_f is simply a sum of fibres and horizontal sections with coefficients $\leq q-1$, and hence, this case is simple. We may assume now that $d > 0$. Then π admits a unique negative section curve C. Note that $f^{-1}(C) = C$. Then we have the decomposition by effective $\mathbb{Q}$-divisors

$$\frac{R_f}{q-1} = C + V + H,$$

where V has support in the fibres of π and each component of H dominates Y. Clearly, C is not a component of H and each component of V has the coefficient ≤ 1. Fix a point $y \in Y$. Let $F := \pi^{-1}(y)$. For the restriction

$$f|_F : F \cong \mathbb{P}^1 \to f(F) \cong \mathbb{P}^1,$$

we have $\frac{R_{f|_F}}{q-1} = \frac{R_f}{q-1}|_F = C|_F + H|_F$. Note that $\frac{R_f}{q-1} \cdot F = 2$, each component of $\frac{R_{f|_F}}{q-1}$ has coefficient ≤ 1, and $C|_F$ is a reduced point. Consider the local intersection number at $x \in F$. If $x = C \cap F$, then $(H \cdot F)_x = 0$, and hence

$$((C+H) \cdot F)_x = 1.$$

If $x \neq C \cap F$, then

$$((C+H) \cdot F)_x \leq (C+H) \cdot F - (C \cdot F)_{C \cap F} = \frac{R_f}{q-1} \cdot F - 1 = 1.$$

In particular, we have $(X, F + C + H)$ is lc near F by [11,Corollary 5.57]. Note that $\frac{R_{f|_F}}{q-1} \leq F + C + H$ near F. So $(X, \frac{R_{f|_F}}{q-1})$ is lc. ■

We give some partial results on the $\mathbb{P}^2$ case that is enough for the proof of Theorem 1.2.

Theorem 4.2. Let $f : X \cong \mathbb{P}^2 \to X$ be a q-polarized endomorphism. Suppose $T_f \neq \emptyset$. Then $(X, \frac{R_f}{q-1})$ is log Calabi–Yau after iteration.

Proof. Write $\frac{R_f}{q-1} = T_f + \Delta_f$. Note that Δ_f and T_f have no common components. By Theorem 3.1, we may assume that Δ_f has coefficient < 1 after sufficient iteration of f. Then the non-klt locus $\mathrm{Nklt}(X, \frac{R_f}{q-1}) = T_f \cup S$ where S is a finite set of points outside T_f.

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We first show that the non-lc locus $\mathrm{Nlc}(X, \frac{R_f}{q-1}) \cap T_f = \emptyset$. In particular, if $S = \emptyset$, then $(X, \frac{R_f}{q-1})$ is lc. Suppose the contrary, and let $x \in \mathrm{Nlc}(X, \frac{R_f}{q-1}) \cap T_f$. Let C be an irreducible component of T_f containing x. By [9], C is a line. After iteration, we may assume $f^{-1}(C) = C$. Consider the log ramification divisor formula,

$$(K_X + C) = f^*(K_X + C) + R_f - (q-1)C.$$

Apply adjunction on C, we have

$$K_C = (f|_C)^* K_C + (R_f - (q-1)C)|_C = (f|_C)^* K_C + R_{f|_C}.$$

So $(R_f - (q-1)C)|_C = R_{f|_C}$. Note that $\frac{R_{f|_C}}{q-1}$ has coefficient ≤ 1. Then $((\frac{R_f}{q-1} - C) \cdot C)_x \leq 1$. By inverse of adjunction (cf. [11, Corollary 5.57]), $(X, \frac{R_f}{q-1})$ is lc near x, a contradiction.

Assume now that $S \neq \emptyset$. Then $\mathrm{Nklt}(X, \frac{R_f}{q-1})$ is not connected. By [10, Theorem 1.2], $(X, \frac{R_f}{q-1})$ is plt, a contradiction. ∎

Corollary 4.3. Let $f : X \cong \mathbb{P}^2 \to X$ be a q-polarized endomorphism. Suppose there is a birational morphism $\pi : W \to X$ from a normal projective surface W such that $\pi \circ h = f \circ \pi$ for some surjective endomorphism $h : W \to W$. Suppose further either the exceptional locus of W is reducible or W has more than one negative curves. Then $T_f \neq \emptyset$ and hence $(X, \frac{R_f}{q-1})$ is log Calabi–Yau after iteration.

Proof. By Proposition 2.7, W is of Fano type and hence $\mathbb{Q}$-factorial (cf. [11, Proposition 4.11]). By [14, Lemma 4.3], the negative curves on W are h^{-1}-periodic. Note that the existence of negative curve on W not contracted by π will cause $T_f \neq \emptyset$. So we may always assume that the exceptional locus of W is reducible.

Let $E = \sum_{i=1}^{n} E_i$ be the reduced π-exceptional divisor with $n > 1$. Write $K_W = \pi^* K_X + \sum_{i=1}^{n} a_i E_i$ where $a_i > 0$ since X is smooth. By the negativity lemma, K_W is not π-nef. So we can run h-equivariant (after iteration) relative minimal model program of W over X that finally ends up with X. Denote by $\tau : W_1 \to X$ the last step and $\sigma : W \to W_1$ the composition of the previous steps. Note that W_1 is of Fano type and $\rho(W_1) = 2$. Denote by $p : W_1 \to Y$ be the contraction induced by the extremal ray different with τ. If p is birational, then the exceptional curve of p is not contracted by τ and hence $T_f \neq \emptyset$. So we may assume that $p : W_1 \to Y \cong \mathbb{P}^1$ is a $\mathbb{P}^1$-fibration. Suppose that E_1 is contracted by σ. Then $p(\sigma(E_1))$ is an $(h|_Y)^{-1}$-invariant point on Y (cf. [3, Lemma 7.5]).

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Therefore, $p^{-1}(p(\sigma(E_1)))$ is an $(h|_{W_1})^{-1}$-invariant curve that is not contracted by τ and hence $T_f \neq \emptyset$. Finally, we apply Theorem 4.2. ■

5 Anti-Canonical Divisor

Throughout this section, we use the following setting.

5.1. Let $f : X \to X$ be a q-polarized endomorphism of a smooth projective threefold X admitting an (f-equivariant) Fano contraction $\pi : X \to Y$ with Y being an elliptic curve and general fibres being $\mathbb{P}^2$. In this setting, $\rho(X) = 2$ and we may further assume $f^*|_{\mathrm{N}^1(X)} = q\,\mathrm{id}$. Denote by

$$\phi : X \dashrightarrow Z$$

the Chow reduction of the Iitaka fibration of $-K_X$, which is f-equivariant by [14, Theorem 7.8].

Proposition 5.2. Suppose there is another fibration $\tau : X \to V$ different with π with $0 < \dim(V) < 3$. Then $-K_X$ is semi-ample. In particular, X is of Calabi–Yau type.

Proof. By the assumption of the setting 1, $\rho(V) = 1$ because $\rho(X) = 2$ and $-K_X$ is not ample because otherwise X is Fano and hence rationally connected contradicting with the setting of a Fano fibration over an elliptic curve. Since π and τ are two different fibrations to positive lower-dimensional varieties, the two extremal rays of $\mathrm{Nef}(X)$ are generated by the pullbacks of ample divisors on Y and V that are not big. Hence, $\mathrm{Nef}(X) = \mathrm{PE}^1(X)$. Note that $-K_X$ is not ample but effective and π-ample. So $-K_X \equiv \phi^*H$ for some ample $\mathbb{Q}$-divisor H on V. By Bertini's theorem, we may assume (X, ϕ^*H) is lc (even terminal) after a suitable choice of H. By the abundance, $-K_X \sim_{\mathbb{Q}} \phi^*H$ and we are done. ■

Theorem 5.3. Assume further $\kappa(X, -K_X) < 3$, that is, $-K_X$ is not big. Then X is of Calabi–Yau type.

Proof. By Proposition 2.5, we may assume $\kappa(X, -K_X) > 0$. By Proposition 5.2, we may assume ϕ is not well defined. Let W be the normalization of the graph of ϕ. Denote by $h : W \to W$ the lifting of f and $\sigma : W \to X$ the induced birational morphism. Since X is smooth, the exceptional locus of σ is of pure codimension one in W (cf. [11, Corollary 2.63]). We denote it by $E := \sum_{i=1}^{n} E_i$ with $n > 0$. After iteration, we may assume E_i is h^{-1}-invariant. Note that the elliptic curve Y admits no g^{-1}-periodic points. So $\pi(\sigma(E_i)) = Y$.

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Let $W_y := (\pi \circ \sigma)^{-1}(y)$ be an h-invariant general fibre after iteration (cf. [5, Theorem 5.1]). Denote by $X_y := \pi^{-1}(y) \cong \mathbb{P}^2$. If E is reducible, that is, $n > 1$, then $E \cap W_y$ is also reducible. By Corollary 4.3, $(X_y, \frac{R_{f|_{X_y}}}{q-1})$ is log Calabi–Yau after iteration of f. By Proposition 3.4, $(X, \frac{R_f}{q-1})$ is log Calabi–Yau after iteration of f. So we may assume E is irreducible. Note that E is $\mathbb{Q}$-Cartier (cf. [11, Lemma 2.62]). Then W is $\mathbb{Q}$-factorial. By [21, Theorem 1.3], (W, E) is lc. Since $W \backslash E$ is smooth, W is klt.

Write $K_W = \sigma^* K_X + aE$ with $a > 0$ since X is smooth. Then,

$$0 \leq \kappa(W, -K_W) \leq \kappa(W, -\sigma^* K_X) = \kappa(X, -K_X),$$

where the 1st inequality follows from Theorem 2.6 and the last equality follows from [19, Theorem 5.13]. Let $\phi_W : W \dashrightarrow V$ be the Chow reduction of the Iitaka fibration of $-K_W$. So $\dim(V) < 3$. Suppose ϕ_W is not well defined. Let $\widetilde{W}$ be the normalization of the graph of ϕ_W and $\widetilde{\sigma} : \widetilde{W} \to W$ the induced birational morphism. Since W is $\mathbb{Q}$-factorial, the exceptional locus of $\widetilde{\sigma}$ is of pure codimension one in $\widetilde{W}$ (cf. [11, Corollary 2.63]) and each irreducible component dominates Y by a similar argument. In particular, the restricted birational morphism $\widetilde{W}_y \to X_y$ has reducible exceptional locus. Then we are done by Corollary 4.3 and Proposition 3.4. So we may assume that ϕ_W is well defined, $\rho(W_y) = 2$ and W_y has only one negative curve $C = E \cap W_y$, that is, the exceptional curve of $W_y \to X_y$. We give more information on W_y. By Proposition 2.7 and [11, Proposition 4.11], W_y is of Fano type and $\mathbb{Q}$-factorial. Since W_y has only one negative curve (which is also K_{W_y}-negative since X is smooth), another contraction of W_y is a $\mathbb{P}^1$-fibration over $\mathbb{P}^1$. In particular, $-K_{W_y}$ is ample.

If $\dim(V) = 0$, then (W, T_h) is log Calabi–Yau by Proposition 2.5. If $\dim(V) = 1$, then ϕ_W is just the Iitaka fibration of $-K_W$, and hence $-K_W$ is semi-ample. In both cases, W and hence X are of Calabi–Yau type by Lemma 2.2.

We are left with $\dim(V) = 2$. Note that $-K_W|_{W_y} = -K_{W_y}$ is ample. So the restriction $\phi_W|_{W_y} : W_y \to V$ is surjective. We first assume that the induced map $X_y \dashrightarrow V$ is not well defined. By the rigidity lemma (cf. [4, Lemma 1.15]), $\phi_W|_{W_y}$ contracts no curves. Then $\phi_W|_{W_y} : W_y \to V$ is finite surjective. By [11, Lemma 5.16 and Proposition 5.20], V is klt and $\mathbb{Q}$-factorial. Let $C_V := \phi_W|_{W_y}(C)$. By the projection formula, C_V is also a negative curve. Then, we have an $h|_V$-equivariant divisorial contraction $V \to V_1$ that contracts C_V. By the rigidity lemma again, the induced rational map $X_y \dashrightarrow V_1$ is then well defined. If the map $X_y \dashrightarrow V$ is already well defined, then we simply identify V_1 with V. Consider the induced f-equivariant dominant rational map $\phi' : X \dashrightarrow V_1$. Note that indeterminant locus of ϕ' is f^{-1}-invariant that does not dominate Y since

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ϕ' is well defined near X_Y. By [3, Lemma 7.5], ϕ' is well defined and we are done by Proposition 5.2. ■

6 Proof of Theorem 1.2

We refer to [1] for well-known facts on vector bundles over elliptic curves. By $\mathcal{F}_n$, we mean the unique indecomposable rank n vector bundle with nontrivial global sections over an elliptic curve. The following lemmas will be used later.

Lemma 6.1. Let $S \cong \mathbb{P}_Y(\mathcal{F}_2)$ where Y is an elliptic curve and $\mathcal{F}_2$ is the unique indecomposable rank 2 vector bundle with nontrivial global sections. Then S admits no polarized endomorphism.

Proof. Suppose the contrary that there is a q-polarized endomorphism $f : S \to S$. We may assume $f^*|_{\mathrm{N}^1(S)} = q\,\mathrm{id}$. Note that $\pi : S \to Y$ has a unique section C with $C^2 = 0$ and $-K_S \sim 2C$. Then $\mathcal{O}_S(1) \cong \mathcal{O}_S(C)$ and $\pi_*\mathcal{O}_S(n) \cong \mathrm{Sym}^n(\mathcal{F}_2) \cong \mathcal{F}_{n+1}$ for $n > 0$ (cf. [1, Theorem 9]). In particular, $h^0(S, nC) = h^0(Y, \mathcal{F}_{n+1}) = 1$ and $\kappa(S, -K_S) = 0$. By Proposition 2.5, we have $\mathrm{Supp}\, R_f = T_f$ and $-K_S \sim_{\mathbb{Q}} T_f$. Note that T_f is reduced. So $T_f \neq 2C$ and hence $\kappa(S, C) > 0$, a contradiction. ■

Proof of Theorem 1.2. **Step 1.** In this step, we reduce our situation to the case when $T_f = \emptyset$, $f^*|_{\mathrm{N}^1(X)} = q\,\mathrm{id}$ and $X \cong \mathbb{P}_Y(\mathcal{E})$ for some rank 3 vector bundle $\mathcal{E}$ on an elliptic curve Y.

By [13, Theorem 1.8], we run f-equivariant minimal model program (after iteration)

$$X = X_1 \dashrightarrow X_2 \dashrightarrow \cdots \dashrightarrow X_r \to Y,$$

where Y is a Q-abelian variety and $g := f|_Y$ that is still polarized. Moreover, each composition $X_i \dashrightarrow Y$ is a well-defined equi-dimensional morphism with all the fibres irreducible and rationally connected. Denote by $\pi : X \to Y$ be the induced composition. If $\dim(Y) = 3$, then $X = Y$ is Q-abelian and hence $(X, \frac{R_f}{q-1}) = (X, 0)$ is log Calabi–Yau. If $\dim(Y) = 0$, then X is of Fano type by Theorem 2.8.

Suppose $\dim(Y) = 2$. By Theorem 2.6, we have $-(K_X + T_f) \sim_{\mathbb{Q}} P$ for some effective $\mathbb{Q}$-divisor P with coefficients ≤ 1 and containing no component of T_f. Let F be a general fibre of π. Since F is general, we may assume that $\mathrm{Supp}\, P \cup T_f$ intersects with F transversally. In particular, $(F, (T_f + P)|_F)$ is lc and $(X, T_f + P)$ is lc over the generic

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point of π. By the canonical bundle formula [6, Theorem 4.1.1], we have

$$0 \equiv K_X + T_f + P = \pi^*(K_Y + B + M).$$

Note that B is effective and M is pseudo-effective and $K_Y \equiv 0$. So $B = 0$ and hence $(X, T_f + P)$ is lc and X is of Calabi–Yau type.

Therefore, we may assume that $\dim(Y) = 1$. Then $f^*|_{\mathrm{N}^1(X)} = q\,\mathrm{id}$ after iteration by [13, Theorem 1.8]. Note that the general fibre F of π is a smooth rational surface. If $\rho(F) > 1$, then we may apply Theorem 4.1 to Proposition 3.4, and hence X is log Calabi–Yau. So we may assume $\rho(F) = 1$ and hence $F \cong \mathbb{P}^2$. Since X is smooth, the 1st step $X_1 \dashrightarrow X_2$ is either a divisorial contraction of a Fano contraction by [16, Theorem 3.3]. If it is divisorial, then the exceptional divisor E is disjoint with the general fibre F since now $F \cong \mathbb{P}^2$, and hence $\pi(E)$ is a point in Y. However, after iteration, $f^{-1}(E) = E$, and hence $g^{-1}(\pi(E)) = \pi(E)$ by [3, Lemma 7.5], a contradiction by [13, Lemma 4.7]. So $X_1 \dashrightarrow X_2$ is a Fano contraction over Y. Again, because $F \cong \mathbb{P}^2$, this is possible only when $X_2 = Y$. In particular, $\rho(X) = 2$ and π is a Fano contraction. By [16, Theorem 3.5] and since the Brauer group of the elliptic curve Y is trivial, $X \cong \mathbb{P}_Y(\mathcal{E})$ for some rank 3 vector bundle $\mathcal{E}$ on Y.

Suppose $f^{-1}(P) = P$ for some prime divisor P. Then P dominates Y since $f|_Y$ is étale and polarized. Let F be an f-invariant (after iteration) fibre of π. Then $D \cap F \subseteq T_{f|_F}$. By Theorem 4.2 and Proposition 3.4, $(X, \frac{R_f}{q-1})$ is log Calabi–Yau. So we may assume $T_f = \emptyset$.

Step 2. In this step, we reduce our situation to the case when $\mathcal{E} \cong \mathcal{F}_2 \oplus \mathcal{L}$ for some line bundle $\mathcal{L}$.

Note that $f|_Y$ is polarized and has Zariski dense periodic points. So after iteration and choosing some fixed point as the identity element of Y, we may assume $f|_Y$ is an isogeny. In particular, $f|_Y$ commutes with any multiplication map. By Lemma 2.3, we can always replace X by base change of multiplication map (by 3) on Y. So we may assume $3|\deg \mathcal{E}$. By Theorem 5.2, we may further assume $-K_X$ is big. Consider the following two cases.

Suppose $\mathcal{E} \cong \mathcal{L}_1 \oplus \mathcal{L}_2 \oplus \mathcal{L}_3$ where $\mathcal{L}_1 \cong \mathcal{O}_Y$. Let D_1 be the divisor determined by the projection $\mathcal{E} \to \mathcal{L}_2 \oplus \mathcal{L}_3$ and define D_2 and D_3 similarly. Consider the exact sequence

$$0 \to \mathcal{O}_X(1) \otimes \mathcal{O}_X(-D_1) \to \mathcal{O}_X(1) \to \mathcal{O}_X(1) \otimes \mathcal{O}_{D_1} \to 0.$$

Taking π_*, we have

$$0 \to \pi_*(\mathcal{O}_X(1) \otimes \mathcal{O}_X(-D_1)) \to \mathcal{E} \to \pi_*\mathcal{O}_{D_1}(1) = \mathcal{L}_2 \oplus \mathcal{L}_3 \to 0$$

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with 0 on the right because $R^1\pi_*(\mathcal{O}_X(1)\otimes\mathcal{O}_X(-D_1))=0$ since all the fibres are $\mathbb{P}^2$. By the projection formula, we have $\mathcal{O}_X(1)\cong\mathcal{O}_X(D_1)$, and similarly, $D_1\sim D_2+\pi^*c_1(\mathcal{L}_2)$ and $D_1\sim D_3+\pi^*c_1(\mathcal{L}_3)$. So, by the relative Euler sequence, we have

$$-K_X\sim 3D_1-\pi^*(c_1(\mathcal{E}))=D_1+D_2+D_3.$$

It is easy to see that $D_1+D_2+D_3$ has simple normal crossing. In particular, $(X,D_1+D_2+D_3)$ is a log Calabi–Yau pair.

Suppose $\mathcal{E}$ is indecomposable. Since $3|\deg\mathcal{E}$, we have $\mathcal{E}\cong\mathcal{F}_3\otimes\mathcal{L}=0$ for some line bundle $\mathcal{L}$ (cf. [1, Theorem 5]). Therefore, we may assume $\mathcal{E}\cong\mathcal{F}_3$. However, $-K_X$ is then nef but not big, a contradiction with our assumption.

Now we may assume $\mathcal{E}\cong\mathcal{F}\oplus\mathcal{L}$ where $\mathcal{F}$ is an indecomposable rank 2 vector bundle and $\mathcal{L}$ is a line bundle. Note that we may also assume that $\mathcal{F}$ does not split after taking pullback of the multiplication map (by 2) of Y. So we may assume that $\mathcal{F}\cong\mathcal{F}_2$.

Step 3. In this step, we show that $\deg\mathcal{L}<0$. Note that there is a non-splitting exact sequence

$$0\to\mathcal{O}_Y\to\mathcal{E}\to\mathcal{O}_Y\oplus\mathcal{L}\to 0,$$

which is induced by the non-splitting exact sequence of $\mathcal{F}_2$. Let D be the divisor determined by $\mathcal{E}\to\mathcal{O}_Y\oplus\mathcal{L}$. Then $\mathcal{O}_X(1)\cong\mathcal{O}_X(D)$. Let D' be the divisor determined by the natural projection $\mathcal{E}\to\mathcal{F}_2$. Then $D\sim D'+\pi^*c_1(\mathcal{L})$ and we have

$$-K_X\sim 3D-\pi^*c_1(\mathcal{L})\sim 2D+D'.$$

Note that $D|_D\sim C$ where C is the section curve determined by $\mathcal{O}_Y\oplus\mathcal{L}\to\mathcal{L}$. Then we see that $-K_X$ is big if and only if $\deg\mathcal{L}\neq 0$. By Lemma 6.1, D' is not f-periodic. Note that $f^*D'\equiv qD'$ and $f^{-1}(D')\neq D'$. So $D'|_{D'}$ is pseudo-effective and hence nef on D'. Then D' is nef. Note that $-K_X$ is not nef because otherwise X is of Fano type. So D is not nef and hence $\deg\mathcal{L}<0$.

Step 4. In this step, we show that $\kappa(X,D)=0$. Note that

$$\pi_*(\mathcal{O}_X(n))\cong\mathrm{Sym}^n(\mathcal{E})\cong\bigoplus_{i=0}^{n}\mathcal{F}_{i+1}\otimes\mathcal{L}^{\otimes(n-i)}$$

for $n\geq 0$. Then,

$$h^0(X,nD)=\sum_{i=0}^{n}h^0(Y,\mathrm{Sym}^i(\mathcal{F}_2)\otimes\mathcal{L}^{\otimes(n-i)})=h^0(Y,\mathcal{F}_{n+1})=1$$

by noticing that $h^0(Y,\mathcal{F}_{i+1}\otimes\mathcal{L}^{\otimes(n-i)})=0$ for any $i<n$. So $\kappa(X,D)=0$.

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End of the proof. By Proposition 2.4, D is f^{-1}-periodic. However, this contradicts the reduction we obtained in Step 1. ■

Remark 6.2. When $X \cong \mathbb{P}_Y(\mathcal{F}_2 \oplus \mathcal{L})$ for some line bundle $\mathcal{L}$ of negative degree, we do not know whether X is of Calabi–Yau type or not. What we proved above is that either X is of Calabi–Yau type or X admits no polarized endomorphism. Therefore, it will be very interesting to study the following question.

Question 6.3. Let $X \cong \mathbb{P}_Y(\mathcal{E})$ be a projective bundle over an elliptic curve Y.

(1) When will X be of Calabi–Yau type?
(2) When will X admit a polarized endomorphism?

Funding

This work was supported in part by Science and Technology Commission of Shanghai Municipality [22DZ2229014]; and the Research Fellowship of KIAS [MG075501].

Acknowledgments

The author would like to thank professor Paolo Cascini for the warm hospitality and inspiring discussion during visiting Imperial Colledge London in 2018. The author would like to thank Doctor Guolei Zhong and the referee for valuable suggestions to improve this paper.

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